

다변수미적분 Black Half Test 01. 정답 및 해설

1)

$$f_{xy}(0,0) = \lim_{y \rightarrow 0} \frac{5x^2 - 3y^2}{x^2 + y^2} \Big|_{x=0} = -3$$

정답 : ④

2) $f(3, 1) = 1$ 이고

$$f_x(x, y) = 4(x-2y)^3 \Rightarrow f_x(3, 1) = 4,$$

$$f_y(x, y) = 4(x-2y)^3 \times (-2) \Rightarrow f_y(3, 1) = -8$$

이고 선형 근사함수를 $L(x, y)$ 라고 할 때,

$$L(x, y) = f(3, 1) + f_x(3, 1)(x-3) + f_y(3, 1)(y-1) \\ = 1 + 4(x-3) - 8(y-1)$$

이다. 따라서 $f(2.9, 1.2)$ 의 근삿값은

$$L(2.9, 1.2) = 1 + 4(-0.1) - 8(0.2) \\ = 1 - 0.4 - 1.6 = -1 \text{ 이다.}$$

[다른풀이]

$$df = f_x dx + f_y dy \text{ 이고}$$

$$f_x(3, 1) = 4, f_y(3, 1) = -8, dx = -0.1, dy = 0.2 \text{ 이므로}$$

$$df = -0.4 - 1.6 = -2 \text{ 이다.}$$

$$\text{따라서 } f(2.9, 1.2) \approx f(3, 1) + df = 1 - 2 = -1 \text{ 이다.}$$

정답 : ①

3) $(x, y) = (0, 1)$ 일 때 $z = 1$ 이고

$$f(x, y, z) = e^{xyz} + xy^2z + \ln(z+xy) \text{ 이라 할 때,}$$

음함수 미분법에 의하여

$$\frac{\partial z}{\partial x} \Big|_{(x,y,z)=(0,1,1)} \\ = - \frac{2xyz e^{xyz} + y^2z + \frac{y}{z+xy}}{x^2y e^{xyz} + xy^2 + \frac{1}{z+xy}} \Big|_{(x,y,z)=(0,1,1)} \\ = - \frac{1+1}{1} = -2$$

4)

$$(\neg) \sum_{n=1}^{\infty} \frac{\cos\left(\frac{n\pi}{3}\right)}{n!} : \text{수렴 (디리클레판정)}$$

$$(\cup) \sum_{n=1}^{\infty} (\tan^{-1} n)^n : \text{발산 } (n \text{ 승근 판정 } \frac{\pi}{2} > 1)$$

$$(\cap) \sum_{n=1}^{\infty} (-1)^n \cos\left(\frac{1}{n^2}\right) : \text{발산 (발산판정)}$$

$$(\exists) \sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2} : \text{수렴 } (n \text{ 승근 판정 } e^{-1} < 1)$$

5)

$$f(x) = \sum_{n=1}^{\infty} \frac{x^n}{3^n n(n+1)} \text{ 에서}$$

$$f'(x) = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{3^n n(n+1)} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{3^n (n+1)} \text{ 이고}$$

$$f'(2) = \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n (n+1)} \\ = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{(n+1)} \left(\frac{2}{3}\right)^n \\ = \frac{3}{4} \sum_{n=1}^{\infty} \frac{1}{(n+1)} \left(\frac{2}{3}\right)^{n+1} \\ = \frac{3}{4} \sum_{n=2}^{\infty} \frac{1}{n} \left(\frac{2}{3}\right)^n$$

정답 : ②

$$= \frac{3}{4} \left(-\frac{2}{3} - \ln\left(1 - \frac{2}{3}\right) \right) \left(\cdots - \ln(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n} \right) \\ = -\frac{1}{2} + \frac{3}{4} \ln 3$$

정답 : ④

6)

$$f(x) = \int_0^x \tan^{-1}(3t^3) dt \text{ 에서 } f'(x) = \tan^{-1}(3x^3) \text{ 이고}$$

$$f'(x) = \tan^{-1}(3x^3) = (3x^3) - \frac{1}{3}(3x^3)^3 + \frac{1}{5}(3x^3)^5 \cdots \\ = 3x^3 - 9x^9 + \frac{3^5}{5}x^{15} \cdots \text{ 이다.}$$

$$\text{따라서 } f^{(10)}(0) = -9 \times 9! \text{ 이다.}$$

정답 : ④

7)

$$D_u f(0,0) = \lim_{h \rightarrow 0} \frac{f(u_1 h, u_2 h) - f(0,0)}{h} \\ = \lim_{h \rightarrow 0} \frac{\frac{u_1^2 h^2 - 3u_2^2 h^2}{u_1^2 h^2 + u_2^2 h^2}}{h} \\ = \lim_{h \rightarrow 0} \frac{u_1^2 - 3u_2^2}{h(u_1^2 + u_2^2)}$$

여기서 이 극한값이 존재하려면 $u_1^2 = 3u_2^2$ 을 만족하면 된다.

$$\therefore u = (u_1, u_2) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right) \quad (\because u_1 > 0, u_2 > 0)$$

따라서 u 와 x 축의 양의 방향과 이루는 각은 $\frac{\pi}{6}$ 이다.

정답 : ④

8)

$$h(x) = \sinh \left(x^2 \int_{x^2}^{x^2-2x+2} e^{t^3} dt \right) \\ \Rightarrow h'(x) = \cosh \left(x^2 \int_{x^2}^{x^2-2x+2} e^{t^3} dt \right) \\ \times \left[2x \int_{x^2}^{x^2-2x+2} e^{t^3} dt + x^2 \left\{ \int_{x^2}^{x^2-2x+2} t^3 e^{t^3} dt \right. \right. \\ \left. \left. + (2x-2)e^{(x^2-2x+2)^3} - 2xe^{x^3} \right\} \right] \\ \Rightarrow h'(1) = \cosh 0 \cdot (-2e) = -2e$$

정답 : ①

9) $f(tx, ty) = t^{100} f(x, y)$ 에서 양변을 x 로 편미분하면

$$f_x(tx, ty) t = t^{100} f_x(x, y) \text{ 이고 } f_x(tx, ty) = t^{99} f_x(x, y) \text{ 이다.}$$

$f(tx, ty) = t^{100} f(x, y)$ 에서 양변을 y 로 편미분하면

$$f_y(tx, ty) t = t^{100} f_y(x, y) \text{ 이고 } f_y(tx, ty) = t^{99} f_y(x, y) \text{ 이다.}$$

따라서 $f_x(tx, ty) + f_y(tx, ty) = t^{99} [f_x(x, y) + f_y(x, y)]$ 이다.

정답 : ①

10)

$$2+3t = x, 3+4t = y \text{ 이라고 하면}$$

$$t = 1 \text{ 일 때 } (x, y) = (5, 7) \text{ 이고,}$$

$$g'(t) = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

$$= 3 \frac{\partial f}{\partial x} + 4 \frac{\partial f}{\partial y} = (f_x, f_y) \cdot (3, 4) \text{ 이다.}$$

이때 $D_u f(5, 7) = 3$ 에 의하여

$$D_u f(5, 7) = (f_x(5, 7), f_y(5, 7)) \cdot \frac{1}{5} (3, 4)$$

$$= \frac{1}{5} (3f_x(5, 7) + 4f_y(5, 7)) = 3$$

$$\Leftrightarrow 3f_x(5, 7) + 4f_y(5, 7) = 15 \text{ 이다.}$$

$$\therefore g'(1) = (f_x(5, 7), f_y(5, 7)) \cdot (3, 4) = 15$$

11)

남동쪽 방향의 단위벡터는 $u = \frac{1}{\sqrt{2}}(1, -1)$ 이다.

$$\nabla f(60, 40) = (-0.6, -0.8) \text{ 이므로}$$

$$\text{변화율은 } \nabla f(60, 40) \cdot \frac{1}{\sqrt{2}}(1, -1) = \frac{\sqrt{2}}{10} \text{ 이다.}$$

정답 : ④

$$\begin{aligned} 12) f(x) &= c_0 + c_1x + c_2x^2 + c_3x^3 + \sum_{n=4}^{\infty} c_nx^n \\ &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \sum_{n=4}^{\infty} \frac{f^{(n)}(0)}{n!}x^n \end{aligned}$$

$$\begin{aligned} &\geq 1 + \frac{1}{2!}x^2 - \frac{1}{3!}x^3 + \sum_{n=4}^{\infty} \frac{(n+1)(n+2)(n+3)}{n!}x^n \\ &= 1 + \frac{1}{2!}x^2 - \frac{1}{3!}x^3 + \sum_{n=4}^{\infty} \frac{1}{(n+1)(n+2)(n+3)}x^n \end{aligned}$$

$$\begin{aligned} f(1) &\geq 1 + \frac{1}{2} - \frac{1}{6} + \sum_{n=4}^{\infty} \frac{1}{(n+1)(n+2)(n+3)} \\ &= 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{60} = \frac{60+30-10+1}{60} = \frac{81}{60} \end{aligned}$$

$$\therefore f(1) \geq \frac{81}{60}$$

$$\begin{aligned} \therefore \sum_{n=4}^{\infty} \frac{1}{(n+1)(n+2)(n+3)} &= \frac{1}{2} \sum_{n=4}^{\infty} \frac{(n+3) - (n+1)}{(n+1)(n+2)(n+3)} \\ &= \frac{1}{2} \sum_{n=4}^{\infty} \left(\frac{1}{(n+1)(n+2)} - \frac{1}{(n+2)(n+3)} \right) \\ &= \frac{1}{2} \left\{ \left(\frac{1}{5 \times 6} + \frac{1}{6 \times 7} + \dots \right) - \left(\frac{1}{6 \times 7} + \dots \right) \right\} \\ &= \frac{1}{60} \end{aligned}$$

13)

ㄱ.

(i) x 축을 따라 원점으로 접근할 때 $\lim_{x \rightarrow 0} \frac{0}{x^2} = 0$

(ii) $x = y^2$ 축을 따라 원점으로 접근할 때

$$\lim_{y \rightarrow 0} \frac{y^4 \cos y}{2y^4} = \frac{1}{2}$$

따라서 극한값은 존재하지 않는다.

$$\therefore -|x^2y| \leq x^2y \sin\left(\frac{1}{x^2+y^2}\right) \leq |x^2y| \text{ 이므로}$$

$$\text{샌드위치정리에 의해 } \lim_{(x,y) \rightarrow (0,0)} x^2y \sin\left(\frac{1}{x^2+y^2}\right) = 0$$

$$\begin{aligned} \therefore \lim_{(x,y) \rightarrow (0,0)} \frac{x \sin^{-1}y}{\sqrt{x^2+y^2}} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2+y^2}} \cdot \frac{\sin^{-1}y}{y} = 0 \times 1 = 0 \end{aligned}$$

14)

$$P = \frac{x+y}{y+z} = 1 + \frac{4\sin w}{u}$$

$$\Rightarrow \frac{\partial P}{\partial u} = -\frac{4\sin w}{u^2}$$

$$\Rightarrow \frac{\partial^2 P}{\partial w \partial u} = -\frac{4\cos w}{u^2}$$

정답 : ②

$$\Rightarrow \frac{\partial^2 P}{\partial w \partial u} \bigg|_{\left(2, \frac{\pi}{3}, \frac{\pi}{2}\right)} = 0$$

정답 : ②

$$15) S_{n+1} = \sum_{k=1}^{n+1} (k+1)(k+2)a_k = \frac{n+4}{n+5} = 1 - \frac{1}{n+5}$$

$$S_n = \sum_{k=1}^n (k+1)(k+2)a_k = \frac{n+3}{n+4} = 1 - \frac{1}{n+4}$$

$$S_{n+1} - S_n = (n+2)(n+3)a_{n+1} = \frac{1}{n+4} - \frac{1}{n+5} = \frac{1}{(n+4)(n+5)}$$

$$\therefore a_{n+1} = \frac{1}{(n+2)(n+3)(n+4)(n+5)} \quad (n \geq 1)$$

$$\therefore a_n = \frac{1}{(n+1)(n+2)(n+3)(n+4)} \quad (n \geq 2)$$

$$\begin{aligned} \sum_{n=1}^{\infty} a_n &= a_1 + \sum_{n=2}^{\infty} a_n = \frac{2}{15} + \frac{1}{3} \sum_{n=2}^{\infty} \frac{3}{(n+1)(n+2)(n+3)(n+4)} \\ &= \frac{2}{15} + \end{aligned}$$

$$\begin{aligned} &\frac{1}{3} \sum_{n=2}^{\infty} \frac{1}{(n+1)(n+2)(n+3)} - \frac{1}{(n+2)(n+3)(n+4)} \\ &= \frac{2}{15} + \frac{1}{3} \left(\frac{1}{3 \times 4 \times 5} - \frac{1}{4 \times 5 \times 6} + \frac{1}{4 \times 5 \times 6} - \dots \right) \\ &= \frac{2}{15} + \frac{1}{180} = \frac{24}{180} + \frac{1}{180} = \frac{25}{180} = \frac{5}{36} \end{aligned}$$

정답 : ④

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